

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

DIAGNOSTIC FIELD GUIDE

Conveyor Belt Drift and Nuisance Trips: A Field Guide for Aggregate and MRF Plants

These are field notes from years of chasing conveyors that won't stay tracked and won't stay running. I'm a controls engineer, not a belt salesman, so I'll tell you what I actually see: most "bad switch" calls are not bad switches. The switch is doing its job. Something upstream moved the belt, loaded it off-center, or let a return idler seize, and the protection device is the messenger. This guide is written for the person standing at the drive with the line down at 2am, deciding whether to reset and run or to lock it out and fix it. When in doubt, the answer is lock it out.

I'm keeping this general enough to cover a crushed-stone plant's overland and stacker belts and a material recovery facility's (MRF) sort-line and infeed belts, because the failure modes rhyme even when the material doesn't.

Before you touch anything: the hazards that kill people on conveyors

Read this even if you skim the rest.

- **Lockout/tagout (LOTO) before any hands-on work.** A conveyor is a machine designed to drag things along. If your hand is near a pinch point (belt-to-pulley, belt-to-idler, return-to-frame) when it starts, you lose the hand or the arm. Verify zero energy: breaker open, locked, tagged, and *tested* by trying to start from the local station after lockout. No exceptions for "I'm just going to clean this one idler."
- **Stored gravity energy on inclines.** This is the one that surprises people. On an inclined belt loaded with material, the belt and load want to run backward when you release the brake or drop the tension. A holdback (backstop) that you're about to service, or a takeup you're about to relax, can let hundreds or thousands of pounds slide. Chock, block, or crib the load and the takeup counterweight before you get under or behind them. Treat the gravity takeup weight as a suspended load — never work under it unblocked.
- **Guarding stays on until LOTO is verified, and goes back on before restart.** Nip guards, return-idler guards, and drive guards exist because people reach in. If you removed a guard to

inspect, it's part of your restore-to-service checklist.

- **Pull-cord and e-stop are safety functions.** People defeat these to stop the production interruptions. You do not defeat them. You repair them. More on that below.
- **Confined spaces: reclaim tunnels, transfer towers, and MRF pits.** Deep tunnels, reclaim tunnels under stockpiles, enclosed transfer towers, and MRF tip/sort pits are commonly permit-required confined spaces, and two hazards stack in them. First, engulfment: a reclaim tunnel under a drawn-down stockpile can bury you if the pile bridges and then lets go — never enter under a live draw. Second, atmosphere: rotting organics in an MRF pit or a poorly ventilated tunnel can go oxygen-deficient or build up toxic gas — hydrogen sulfide (H₂S) from decomposing waste — and any running engine or heater nearby adds carbon monoxide (CO). Before anyone enters, follow your permit-required confined-space program: ventilate, atmospheric-test the space, and wear a bump-tested personal 4-gas monitor (oxygen O₂, lower explosive limit LEL, hydrogen sulfide H₂S, carbon monoxide CO). Note the 4-gas monitor does not see a combustible dust cloud — dust explosion is a separate hazard you control with housekeeping and ignition-source control, not with the meter. Know which of your spaces are permit-required before you send someone in.

The rest of this guide assumes you either have the belt locked out for hands-on work, or you are doing running observation from a safe standing position outside all guards and nip points.

Part 1: Why belts drift (mistracking) and how to correct it without making it worse

Belt tracking is a system. A belt runs off to one side because the forces across its width aren't balanced, and it will keep chasing the imbalance until something stops it — a frame, a drift switch, or the edge shredding on structure. The cardinal rule: **a belt steers off the component it touches first.** Whatever the belt contacts first, as it approaches, is what steers it. That's why you correct tracking working *in the direction of belt travel*, and why yanking one idler over is almost always the wrong move — you've treated a symptom two frames downstream of the real cause.

Here's the honest order of what causes drift, roughly most-common first:

Off-center loading. The number one cause in aggregate and MRF work. If the chute dumps material to one side of the belt centerline, that side is heavier, the belt cups toward it, and it walks. On MRF lines a surge of heavy mixed load (a wet cardboard slug, a chunk of construction-and-demolition debris, C&D) does the same thing intermittently — so the belt tracks fine empty and drifts only under load. Check where the material lands relative to belt center. Fix the chute, the deflector, or the skirt geometry, not the idlers.

Material buildup on idlers and pulleys. Fines, mud, and MRF film/tape build up on return idlers and on the pulley face. A quarter inch of packed buildup on one end of a pulley effectively makes that end a larger diameter, and the belt climbs toward the built-up (larger) side, or away from it depending on

where it packs — either way it steers. This is the single most common *intermittent* tracking problem, and it's why cleaning the return run fixes more tracking calls than adjusting anything. A seized return idler that's worn flat on one side does the same thing.

Seized idlers. A bearing that's locked up stops the roll from turning. The belt now drags across a stationary steel tube. It wears a flat, it grabs unevenly, it heats, and it steers. Seized rolls also throw belt to one side and are a fire risk in dry dusty plants — a locked roll polishing against a moving belt is a real ignition source in a stockpile tunnel. Spin every return idler by hand (LOTO first) and mark the dead ones.

Frame square and level. If the conveyor structure is racked, twisted, or a bent (one of the vertical support frames) has settled, the idlers are no longer perpendicular to belt travel, and the belt walks toward the low/forward corner. This is common after ground settling, forklift hits, and on portable/relocatable plant conveyors that got set up in a hurry. String-line the frame and check that idler frames are square to centerline before you blame anything that moves. A structure problem masquerades as a chronic one-direction drift that no amount of idler adjustment holds.

Idler tilt and troughing idler alignment. Troughing idlers are sometimes deliberately tilted forward (a few degrees) in the direction of travel to aid tracking. Too much tilt accelerates edge and cover wear; the wrong tilt steers the belt. This is a tuning tool, not a first move.

Crowned vs. flat pulleys. A crowned pulley (larger diameter at center) self-centers the belt — the belt climbs to the high center. Tail and takeup pulleys are often crowned for this reason; head/drive pulleys frequently are not, or are lagged flat. If someone swapped a crowned pulley for a flat one during a repair, or the crown wore off a rubber-lagged pulley, you lose the self-centering and the belt wanders. Know which of your pulleys are supposed to be crowned.

The methodical way to correct tracking

Do this with the belt running empty first, then under load, and change one thing at a time. Give the belt several full revolutions to respond before you judge a change — a long overland belt can take minutes to settle. Rushing from adjustment to adjustment chasing the same wander is how people spend a shift and end up worse than they started.

1. **Clean first.** Clean the return run, the pulley faces, and the idlers. Fix the belt cleaner (scraper) if carryback (material that clings to the belt past the discharge and rides the return run back) is feeding the buildup. A shocking number of tracking problems end here.
2. **Fix loading.** Center the load. Repair skirting so material isn't spilling to one side. Check for a plugged or half-plugged chute deflecting flow.
3. **Check the structure.** Square and level the frame. Fix racked bents before touching idlers.
4. **Replace what's dead.** Seized idlers, worn-flat rolls, a flat-spotted crowned pulley — replace them.
5. **Then, and only then, tune with idlers.** Work in the direction of belt travel. To move the belt away from one side, advance (lead) the end of the *carrying* idler on the side the belt is running *toward* —

the belt steers off the idler it reaches first. Adjust the training idler nearest the problem, upstream of where the belt runs off, a small amount, and wait. Self-aligning training idlers can help but they mask root causes; don't install them to avoid fixing loading or structure.

If you find yourself cranking a single idler hard over to hold the belt, stop. You're fighting a cause you haven't found. The belt will tell on you the next time the load surges.

Part 2: The protection devices that pull the interlock — and get blamed

Every one of these exists to stop the belt before something worse happens. When a plant calls them "nuisance trips," nine times out of ten the device caught a real condition and the plant would rather not deal with the condition. The job is to tell a *real* protective trip from a genuine *nuisance* trip, and to fix the right thing.

Belt-drift / misalignment switches. Roller-actuated switches mounted at the belt edge, usually in pairs, near the head and tail. The belt edge contacts the roller; if it pushes far enough, the first stage warns and the second (further) stage trips. A drift switch tripping means *the belt moved that far*. That is real. The fix is Part 1 of this guide — find why it's drifting. The wrong fix, and the one I get called to undo, is bending the switch bracket outward or moving it back so the belt has to travel farther before it trips. Now the belt shreds its edge on structure before the switch ever sees it. You didn't fix drift; you disabled the thing that was protecting a \$40k belt. If the switch trips with the belt visibly centered, *then* suspect the switch: bent actuator arm, corroded pivot, or a bracket knocked in by a chunk of material.

Pull-cord / e-stop (lanyard) switches. The rope running the length of the conveyor that anyone can grab to stop it. **This is a safety function. You never defeat it, widen its tolerance, or jumper it out.** If it's nuisance-tripping, the honest causes are: the rope tension is out of spec (sagged rope in the heat, over-tight rope in the cold — these switches trip on both loss of tension and pull, by design), a corroded switch that's lost its detent, a broken rope, or a rope fouled on structure or ice. Every one of those is a *repair*, not a bypass. Re-tension to the switch's spec, replace the corroded switch, free the rope. Confirm both directions still trip and that the reset requires the switch itself to be reset (a proper pull-cord latches and won't clear until you reset it at the switch — that's a feature, verify it still works). If a controls "fix" you inherited involves a jumper across a pull-cord terminal, that comes out today and the reason gets written up.

Zero-speed / underspeed (slip) detection. A speed sensor at the tail or a bend pulley watching that the belt is actually moving at commanded speed. It trips on belt slip, a broken/mistracked belt, a sheared drive coupling, or a stalled belt under a plugged chute. A zero-speed trip is *usually real and usually important* — belt slip on a drive pulley polishes the lagging and can start a fire; a slipping drive on an incline can let the loaded belt run backward against the holdback (backstop), which then catches with a shock load. Before you blame the sensor: check drive-pulley lagging (worn smooth = slip), takeup tension (low tension = slip), and target/sensor gap and buildup on the sensor. The wrong

fix is widening the underspeed threshold so it ignores slip. The right fix is restoring tension and lagging so the belt doesn't slip. Only after you've confirmed the belt genuinely runs at speed should you suspect a fouled sensor, a packed-with-fines target, or a failed proximity switch.

Plugged-chute / blocked-chute switches. Tilt/plumb-bob, capacitance, or paddle sensors in transfer chutes that detect a backing-up plug. In a plugged-chute trip the honest question is: *is the chute actually plugging?* On aggregate transfers, wet fines and clay pack chutes; on MRF lines, film, textiles, and long items bridge and plug. If it's really plugging, the fix is upstream — flow, liner wear, chute geometry, a worn-out belt cleaner feeding carryback into the transfer. If the chute is clear and the switch still trips, then suspect buildup on a capacitance probe (very common in dusty/humid plants — the caked dust reads as material), a stuck paddle, or a tilt switch that got bumped. Do not just crank the sensitivity down until it stops calling; you'll miss the plug that buries a transfer tower.

Belt-rip detectors. Loops or sensor bars across the belt that detect a longitudinal rip, usually near the loading zone where a trapped tramp object (a slab, a spike, a chunk of steel in the MRF stream) can spear the belt and unzip it for hundreds of feet in seconds. A rip-detector trip is the highest-stakes "is this real?" call you'll make. **Assume it's real and inspect the belt.** A rip caught at two feet is a patch; a rip missed becomes a full belt replacement and a tower full of spilled material. False trips do happen — a damaged sensor loop, a splice passing over the sensor, moisture on the electronics — but you earn the right to call it false only after you've walked and inspected the loading-zone belt with it locked out.

Motor overload. The starter/drive overload that trips on sustained overcurrent. On a conveyor, an overload trip is often the *end symptom* of everything above: a seized idler dragging, a plugged chute loading the belt, a slipping-then-grabbing belt, a frozen or overloaded takeup, or simply too much material. It can also be a genuine electrical fault — single-phasing, a failing motor, a loose connection heating up. The tell: does it trip on starting (mechanical drag, undersized start, low voltage) or after running loaded a while (thermal, real overload, cooling problem)? Before resetting an overload more than once, find the load. Repeated overload resets without finding the cause is how motors and couplings die, and how a dragging seized idler becomes a tunnel fire. If it trips instantly on start with no load, suspect electrical (phase loss, shorted winding) and get a meter on it.

Real trip vs. nuisance trip — how to actually tell

The discipline that separates a good troubleshooter from a switch-widener:

- **Does the physical condition exist right now?** Belt off center → drift switch is right. Chute backing up → chute switch is right. Belt not at speed → zero-speed is right. Look before you judge the switch.
- **Does it trip under a specific condition (load surge, wet material, cold morning, one particular product)?** That's the device catching a *real intermittent* condition. Chase the condition, not the device.

- **Does it trip with the belt visibly fine, empty, and clean?** Now you've earned the right to suspect the device — actuator, wiring, corrosion, buildup on the sensor.
- **Log every trip with time, load state, material, weather, and what you found.** Three logged trips with a pattern tell you more than any single reset. A trip log is the cheapest diagnostic tool in the plant and the one nobody keeps.

The wrong fixes, collected in one place so they're easy to refuse: bending or relocating a drift switch so the belt travels farther; jumpering or slackening a pull-cord; widening the underspeed window to ignore slip; turning down chute-switch sensitivity to silence a real plug; disabling a rip detector; and resetting a motor overload on repeat without finding the drag. Every one of these trades a cheap, correct repair for an expensive, dangerous failure later.

Part 3: Corrosive and dusty environments — why good switches go bad

Aggregate plants are dust and grit; MRFs add humidity, rot, and mildly corrosive leachate; anything near a wash plant or a coastal site adds salt and constant moisture. Switches don't fail because they're cheap — they fail because of where they live.

- **Caked dust on capacitance/proximity sensors** reads as material. This is the classic false plugged-chute and false zero-speed trip. Clean the sensor face on a schedule and check the gap; don't chase the setpoint.
- **Corroded pivots and actuator arms** on drift and pull-cord switches stiffen, stick, or lose their return spring. The switch either won't trip or won't reset cleanly. Exercise and lubricate per spec; replace when the detent is gone.
- **Water ingress** into switch enclosures with failed seals corrodes contacts and causes intermittent open circuits that look exactly like nuisance trips. Check that enclosure ratings match the wash-down or weather exposure, and that glands and covers are actually sealed. A cheap switch in the right enclosure beats an expensive one that's full of water.
- **Vibration** backs off actuator setscrews and moves brackets over time, so a switch that was set correctly last year now trips early or late. Re-verify switch positions during PMs, not just after failures.
- **Rodent and material damage** to pull-cord ropes and cabling in tunnels and towers is common and under-inspected. Walk the rope.

If a plant has a "chronic nuisance switch," the switch is usually the right kind of switch in the wrong enclosure, on a corroded pivot, or reading buildup. Fix the environment problem and the "nuisance" stops.

Symptom → likely cause → check (and log)

Symptom	Likely cause (most-likely first)	Check — and log what you find
Belt drifts only under load, fine when empty	Off-center loading; load surge	Watch where material lands vs. center; check skirting/chute deflector. Log: which side, which product, load state.
Belt drifts one direction always, empty or loaded	Racked/settled frame; worn crown on pulley; seized idler	String-line frame for square/level; spin idlers (LOTO); check pulley crown. Log: which frame/bent, which idler.
Belt wander is intermittent, roams both ways	Material buildup on return idlers/pulley; failing belt cleaner (carryback)	Inspect and clean return run and pulley faces; check scraper. Log: buildup location and thickness.
Drift switch trips, belt visibly off center	Real drift — see all drift causes above	Do NOT move the switch. Run Part 1 tracking procedure. Log: which switch, belt offset.
Drift switch trips, belt centered	Bent/corroded actuator; bracket knocked in	Inspect switch arm and pivot; verify bracket position vs. spec. Log: switch condition.
Pull-cord trips with no one pulling	Rope tension out of spec (heat sag/cold tight); corroded switch; fouled/broken rope	Re-tension to switch spec; walk rope for foul/damage; replace corroded switch. NEVER bypass. Log: rope condition, weather.
Zero-speed / underspeed trips	Belt slip (worn lagging, low tension); sheared coupling; plugged chute stalling belt; fouled sensor	Check lagging, takeup tension, coupling; confirm belt actually at speed before blaming sensor. Log: tension reading, lagging condition.
Plugged-chute switch trips	Real chute plug (wet fines/clay, MRF bridging); OR caked buildup on capacitance probe	Look in the chute. If clear, clean/inspect sensor. Do NOT lower sensitivity blindly. Log: plug material or sensor buildup.
Belt-rip detector trips	Tramp object rip at loading zone (assume real); OR damaged sensor loop/splice/moisture	LOTO and inspect loading-zone belt fully before calling it false. Log: rip found or sensor condition.
Motor overload trips after running loaded	Mechanical drag (seized idler, frozen takeup); overloaded belt; slip-and-grab; cooling issue	Find the drag before resetting again. Check idlers, takeup, load, motor cooling. Log: trip vs. load/time.

Symptom	Likely cause (most-likely first)	Check — and log what you find
Motor overload trips instantly on start, no load	Electrical — phase loss, shorted winding, low voltage, undersized start	Meter the motor and supply; do NOT keep resetting. Log: voltage/current readings.
Idler squeals, smokes, or belt has a shiny wear stripe	Seized idler dragging (fire risk in dusty tunnel)	LOTO immediately; replace idler; inspect belt for heat damage. Log: idler location.

10-minute triage: belt down or nuisance-tripping

Work it in this order. Safest-first, most-likely-first.

- 0–1 min — Read the fault.** What tripped it? Get the specific device from the HMI/annunciator or the tripped switch itself (a latched drift or pull-cord switch shows which one). Write down the time and the load state (running loaded, starting, empty).
- 1–2 min — Rip detector or pull-cord? Stop and treat as real.** If it's a belt-rip trip, plan to LOTO and inspect before anything else. If it's a pull-cord, confirm no one is hurt or signaling, then treat as a repair, never a reset-and-run.
- 2–4 min — Look at the belt from a safe standing spot (belt still locked out if you're near it).** Is it off center? By how much and which way? Is a chute backing up? Is material dumping to one side? Your eyes settle most of these before you touch a tool.
- 4–6 min — LOTO and verify zero energy** if you're going hands-on. Test-start to confirm dead. Block the takeup counterweight and chock the load on any incline before getting behind or under anything.
- 6–8 min — Spin and feel.** Spin return and carrying idlers by hand — mark the seized/rough ones. Feel the drive pulley lagging (worn smooth?). Check for buildup on pulley faces and on any capacitance/proximity sensor. Check takeup position (bottomed out, frozen, low tension?).
- 8–9 min — Decide: real or nuisance.** Condition present (off-center, plug, slip, drag, buildup)? It's real — fix the condition. Belt clean, centered, at spec, and the device still calls it? Now inspect the device (actuator, corrosion, wiring, sensor gap).
- 9–10 min — Do the correct fix, one change at a time, and log it.** Clean the return run, center the load, replace the seized idler, re-tension the takeup, re-tension or replace the pull-cord switch, clean the sensor. Restore guards. Re-verify the protection trips correctly before you hand it back. Never widen or bypass a switch to clear a fault.

If the pattern isn't obvious after this, the trip log is your friend — three trips tagged with load, material, and weather will show you the condition the device keeps catching.

A note on getting help

Conveyor faults are some of the worst to hand between vendors, because the belt guy blames controls, the controls guy blames the belt, and the switch keeps tripping in the seam between them. An independent second look tends to pay off most exactly there — on the faults nobody owns — because the honest answer usually isn't in either scope alone: it's a seized idler feeding a drift switch, or a worn cleaner feeding a chute switch. Find the condition, fix the condition, leave the protection alone, and keep the log.

Field notes, not a substitute for your machine's manuals or your plant's safety program. Numbers here are typical/approximate — check tensions, tilt angles, switch tolerances, and reset behavior against your specific equipment's spec and your site's LOTO and confined-space procedures.